ECON 609 - Problem Set 3
Fall 2022

Be sure to include all the material that shows your work and to present your results in a way that is easy to understand.

Kalman Filter vs. Particle Filter

Consider the following linear state-space representation

$$
\begin{aligned}
y_t &= \Psi s_t + u_t \\
s_t &= \Phi s_{t-1} + \epsilon_t
\end{aligned}
$$

where $\mathbb{E}[u_t u_t'] = \Sigma_u$ and $\mathbb{E}[\epsilon_t \epsilon_t'] = \Sigma_\epsilon$. Let the dimension of y_t be 1×1 and s_t be 2×1. Consider the following parameterization

$$
\Psi = \begin{bmatrix} 1 & 0.9 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.5 \end{bmatrix}, \quad \Sigma_u = 0.1^2, \quad \Sigma_\epsilon = \begin{bmatrix} 0.3^2 & 0 \\ 0 & 0.25^2 \end{bmatrix}.
$$

This is the same state-space representation as we saw in Problem Set 1. Given the answer to the Problem Set 1, we have generated $T = 50$ observations by setting the initial s value to be zero. Since this state-space representation is linear and Gaussian, one can evaluate the likelihood and draw the underlying states using Kalman filter. What we will do in this exercise is to compare the performance of Kalman filter and particle filter.

(1) Bootstrap Particle Filter

Code up the bootstrap particle filter. Set the number of particles to $N = 500$. Compute particle approximations for $\mathbb{E}[s_t|Y_{1:t}]$ and $\log p(y_t|Y_{1:t-1})$. Run the particle filter $N_{run} = 50$ times and compute the mean-squared error (across the 50 runs of the PF) of the particle approximations relative to the "truth" produced by the Kalman filter.

(2) Effect of Measurement Error Variance

What happens to the accuracy of the bootstrap particle filters as you reduce the variance of the measurement error Σ_u? Explain.

(3) Effect of Outliers

Now multiply the twenty-fifth observation y_{25} in your sample by a factor of 10. Assess the accuracy of the bootstrap particle filter compared to the Kalman filter.

(4) Conditionally-Optimal Particle Filter

Code up the conditionally-optimal particle filter and repeat the analysis in (1). Compare the accuracy of the bootstrap particle filter and the conditionally-optimal particle filter. By how much can you increase/reduce the number of particles from the conditionally-optimal filter, such that it produces the similar accuracy as the bootstrap filter?